Spring 2013 / EC571

Instructor:Ajay Joshi

Final Exam

Date: May 9, 2013

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	25	
2	25	
3	25	
4	25	
5	25	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours
2. The maximum grade is 100 points
3. Solve any 4 questions. If you solve all 5 questions, then the 4 best scores will be chosen to calculate the final grade.
4. Please write your name on all pages
5. The solutions should be written in the spaces provided on the test sheet
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number
7. This is an open book and open notes exam.
8. Material sharing is not allowed
9. Calculators are allowed
10. Boston University Academic Code of Conduct applies

Question 1

Consider the circuit shown below. Assume a unit clk-to-q delay for the register and a unit delay for the logic block. The registers are positive-edge triggered and have a setup time of one unit. The multiplexer has 2 units of delay.

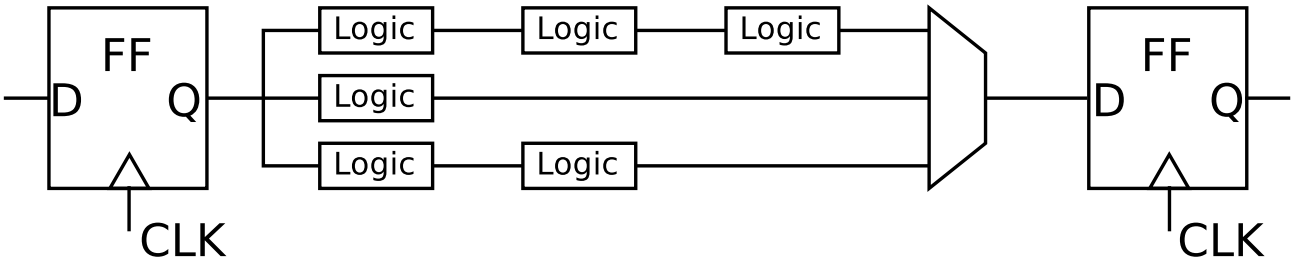


Figure 1

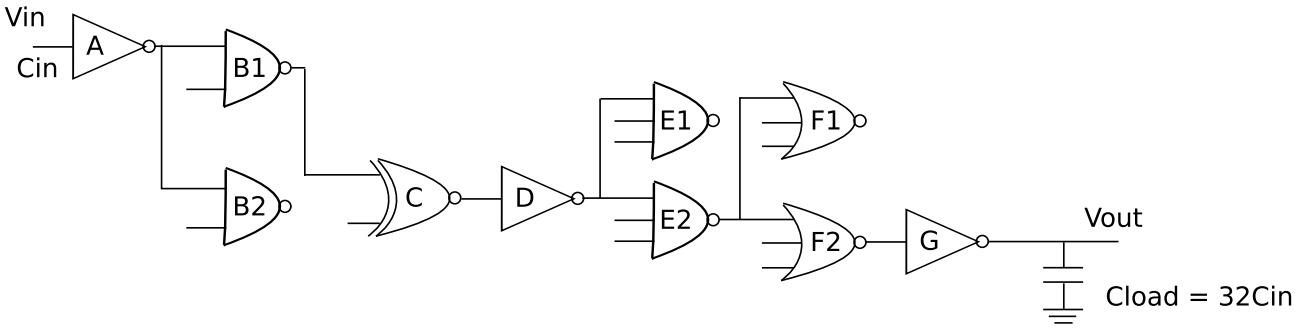
1. Determine the minimum clock period for the above circuit. You can ignore clock skew. (5 points)

2. Determine the maximum hold time for the above circuit. You can ignore clock skew. (5 points)

3. Determine the minimum clock period considering a negative clock skew of 2 time units. (5 points)

4. Assuming a hold time of 1 time unit, what is the maximum positive clock skew that can be tolerated by this circuit? (5 points)

Question 2



Using the logical effort technique, determine the input capacitances for the various gates in the data path from Vin to Vout for the logic design shown above. You can assume that B1 and B2 have the same sizes, E1 and E2 have same sizes and F1 and F2 have the same sizes. (20 points)

Question 3

Design the static complementary CMOS implementation of a 4:1 multiplexer circuit operating at 5 GHz with a load of 50 fF. As part of the design you need to determine the width of all transistors. You can use the following transistor parameters for the design - L = 0.25 micron $R_{sq}(NMOS)$ = 13 Kohm/sq, $R_{sq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron. You can assume that you have both complemented and non-complemented inputs and that input signals have very small rise and fall times. (20 points)

Question 4

Sketch the schematics 2-input XOR functions using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

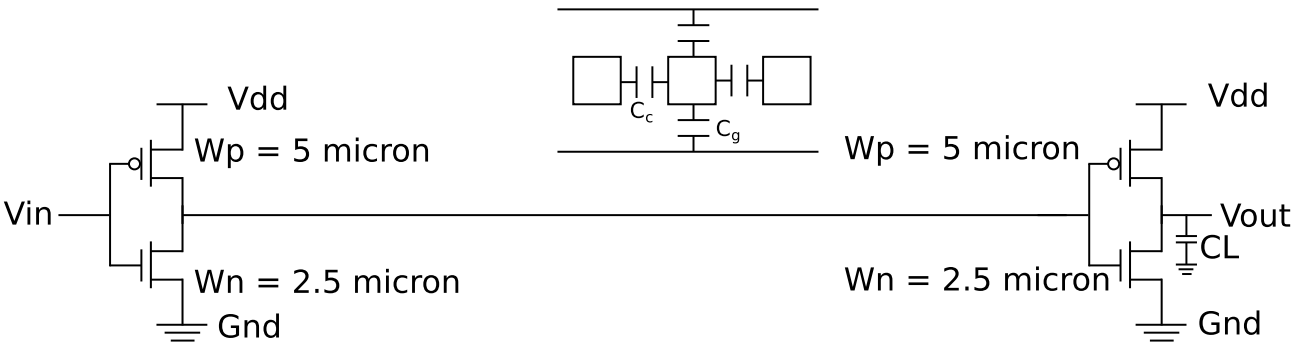
1. Static CMOS. (5 points)

2. Pseudo-nMOS. (5 points)

3. Dual-rail domino. (5 points)

4. Pass transistors. (5 points)

Question 5



The figure above shows a 5 *mm* long interconnect driven by a inverter driver. There is an inverter receiver at the end of the interconnect. Determine the width of the wire such that total delay of the circuit (from Vin to Vout) is 1.5 *nsec* (in the worst case). Calculate the total dynamic energy consumption of the circuit over 500 cycles. Assume 125 $0 \rightarrow 1$ transitions and 125 $1 \rightarrow 0$ transitions. Ignore the input capacitance of the first inverter.
You can assume that the input at Vin changes from 0 to 1 instantaneously. Use a distributed RC model (with a large number of RC elements) for the wire. Use simple parallel plate model for calculating the wire capacitance. Consider ground capacitance as well as coupling capacitance of the wire. You can use the following parameters for the transistors and wires: V_{dd} = 2.5 V, L = 0.25 micron $R_{eq}(NMOS)$ = 13 Kohm/sq, $R_{eq}(PMOS)$ = 26 Kohm/sq, C_g = 1.5 fF/micron, C_d = 2 fF/micron, $Wire_{thickness}$ = 250 micron, $Wire_{length}$ = 5 mm, $Wire_{spacing}$ = = 250 micron, $Dielectric_{thickness}$ = 250 micron, ρ = 1.67 10^{-8} $\Omega - m$, C_L = 50 fF, Dielectric constant = 3.0, ε_0 = 8.85e-12 F/m.

Bonus question

Prof. Joshi is planning on playing 5 different outdoor sports (one on each of the 5 weekdays) this summer. Please suggest 5 different outdoor sports that he can play in and around campus.